Basic & Clinical Pharmacology & Toxicology

Basic & Clinical Pharmacology & Toxicology, 2014, **114**, 174–180 Doi: 10.1111/bcpt.12122

The Role of Pre-junctional D_2-like Receptors Mediating Quinpirole-Induced Inhibition of the Vasodepressor Sensory CGRPergic Out-flow in Pithed Rats

Guadalupe Manrique-Maldonado, Abimael González-Hernández, Alain H. Altamirano-Espinoza, Bruno A. Marichal-Cancino, Inna Ruiz-Salinas and Carlos M. Villalón

Departamento de Farmacobiología, Cinvestav-Coapa, Tlalpan, México D.F., México

(Received 14 March 2013; Accepted 6 August 2013)

Abstract: Calcitonin gene-related peptide (CGRP) released from perivascular sensory nerves plays a role in the regulation of vascular tone. Indeed, electrical stimulation of the perivascular sensory out-flow in pithed rats produces vasodepressor responses, which are mainly mediated by CGRP release. This study investigated the potential role of dopamine D_1-like and D_2-like receptors in the inhibition of these vasodepressor responses. For this purpose, male Wistar pithed rats (pre-treated i.v. with 25 mg/kg gallamine and 2 mg/kg min. hexamethonium) received i.v. continuous infusions of methoxamine (20 μg/kg min.) followed by physiological saline (0.02 ml/min.), the D_1-like receptor agonist SKF-38393 (0.1–1 μg/kg min.) or the D_2-like receptor agonist quinpirole (0.03–10 μg/kg min.). Under these conditions, electrical stimulation (0.56–5.6 Hz; 50 V and 2 ms) of the thoracic spinal cord (T_9–T_{12}) resulted in frequency-dependent vasodepressor responses which were (i) unchanged during the infusions of saline or SKF-38393 and (ii) inhibited during the infusions of quinpirole (except at 0.03 μg/kg min.). Moreover, the inhibition induced by 0.1 μg/kg min. quinpirole (which failed to inhibit the vasodepressor responses elicited by i.v. bolus injections of exogenous α-CGRP; 0.1–1 μg/kg) was (i) unaltered after i.v. treatment with 1 ml/kg of either saline or 5% ascorbic acid and (ii) abolished after 300 μg/kg (i.v.) of the D_2-like receptor antagonists haloperidol or raclopride. These doses of antagonists (enough to completely block D_2-like receptors) essentially failed to modify *per se* the electrically induced vasodepressor responses. In conclusion, our results suggest that quinpirole-induced inhibition of the vasodepressor sensory CGRPergic out-flow is mainly mediated by pre-junctional D_2-like receptors.

It has been widely documented that dopamine regulates a broad range of physiological functions (including cardiovascular homeostasis), and it contributes to blood pressure control due to its peripheral action on the kidney, adrenal glands and vascular tone [1–7]. With the conjunction of structural, transductional and operational (pharmacological) criteria, dopamine receptors can be classified into D_1-like (which includes the D_1 and D_5 subtypes and activates G_s proteins) and D_2-like (which includes the D_2, D_3 and D_4 subtypes and activates $G_{i/o}$ proteins) types [1,2,6]. In the cardiovascular system, D_1-like receptors mediate direct vasodilatation, cardiac stimulation and natriuresis on the renal proximal tubules; whereas activation of pre-junctional D_2-like receptors located on perivascular and cardiac sympathetic nerves results in sympatho-inhibition [1–7].

Regarding resistance blood vessels, these are mainly innervated by sympathetic [8] and primary sensory [9] nerves which modulate the vascular tone. The perivascular sensory nerves are mainly C-fibres originating from the spinal cord and, upon stimulation, cause a non-adrenergic, non-cholinergic (NANC) vasodilatation *via* the release of neuropeptides, primarily calcitonin gene-related peptide (CGRP) [9]. CGRP is predominantly located in sensory neurons (including perivascular nerves), where it is colocalized with other neuropeptides, such as substance P and neurokinin A [10].

Interestingly, Taguchi *et al.* [9] have shown that electrical stimulation of the thoracic (T_9–T_{12}) spinal cord in pithed rats receiving i.v. infusions of hexamethonium and methoxamine caused vasodepressor responses, which are mainly mediated by CGRP release from capsaicin-sensitive primary sensory nerves [9]. More recently, our group has reported that this vasodepressor sensory out-flow can be inhibited by pre-junctional $\alpha_{2A/2C}$ -adrenoceptors [11] as well as by serotonin 5-HT_{1B} [12] and 5-HT_{1F} [13] receptors located on perivascular sensory nerves. Nevertheless, to our knowledge, no study has yet reported whether activation of dopamine receptors can inhibit the vasodepressor sensory CGRPergic out-flow in this model. Only one *in vitro* study [14] has shown that dopamine inhibited the NANC neurotransmission in rat bladder and that this response was partially reversed by haloperidol, implying activation of dopamine receptors. However, this study did not investigate the role of D_1-like and D_2-like receptors on CGRP release. Moreover, as dopamine can also interact with α- and β-adrenoceptors [3], it is not ideal for identifying the pharmacological profile of the receptors involved [5]. On this basis, using selective agonists/antagonists for dopamine receptors, the present study investigated the potential role of D_1-like and D_2-like receptors in the inhibition of the vasodepressor sensory CGRPergic out-flow in pithed rats.

Author for correspondence: Carlos M. Villalón, Departamento de Farmacobiología, Cinvestav-Coapa, Czda. de los Tenorios 235, Col. Granjas-Coapa, Deleg. Tlalpan 14330, México D.F., México (fax (Int.) (52) (55) 5483 2863, e-mail: cvillalon@cinvestav.mx).

Materials and Methods

Animals. Male Wistar normotensive rats (300–350 g) were maintained at a 12-hr light–dark cycle (with light beginning at 07:00 hr) and housed in a special room at constant temperature (22 ± 2°C) and humidity (50%), with food and water *ad libitum*. All animal protocols in this study were approved by our Institutional Ethics Committee (CICUAL-Cinvestav; permission protocol number 507-12) and followed the regulations established by the Mexican Official Norm (NOM-062-ZOO-1999) in accordance with the guide for the Care and Use of Laboratory Animals in the USA [15] and with the ARRIVE guidelines for reporting experiments involving animals [16].

General methods. Experiments were carried out in a total of 108 rats. After the anaesthesia with diethyl ether and cannulation of the trachea, the rats were pithed by inserting a stainless steel rod as reported earlier [11–13,17]. This procedure basically consists of inserting a stainless steel rod through the orbit and *foramen magnum* into the vertebral *foramen*, to avoid the influence of the central nervous system. Then, the animals were artificially ventilated with room air using a model 7025 Ugo Basile pump (56 strokes/min.; stroke volume: 20 ml/kg), as previously established by Kleiman and Radford [18]. After bilateral vagotomy, catheters were placed in (i) the left and right femoral veins for the infusions of agonists and methoxamine, respectively; (ii) the left jugular vein for the infusion of hexamethonium; and (iii) the right jugular vein, for the bolus injections of gallamine or exogenous α-CGRP. In addition, the left carotid artery was connected to a Grass pressure transducer (P23 XL), for recording blood pressure. Both heart rate (measured with a 7P4F tachograph) and blood pressure were recorded simultaneously by a model 7D Grass polygraph (Grass Instrument Co., Quincy, MA, USA).

At this point, the 108 rats were divided into two main sets, so that the effects produced by i.v. continuous infusions of either the vehicle (physiological saline) or the agonists (SKF-38393 or quinpirole) could be investigated on the vasodepressor responses induced by either (i) electrical stimulation of the perivascular (vasodepressor) sensory out-flow (set 1; n = 93) or (ii) i.v. bolus injections of exogenous α-CGRP (set 2; n = 15). The vasodepressor stimulus–response curves (S-R curves) and dose–response curves (D-R curves) elicited by electrical stimulation and exogenous α-CGRP, respectively, were completed after about 50 min., with no change in resting blood pressure. The electrical stimuli (0.56, 1, 1.8, 3.1 and 5.6 Hz), as well as the doses of exogenous α-CGRP (0.1, 0.18, 0.31, 0.56 and 1 μg/kg), were given using a sequential schedule at 5- to 10-min. intervals as previously reported [11–13]. The body temperature of each pithed rat was maintained at 37°C by a lamp and monitored with a rectal thermometer.

Experimental protocols. After the animals had been in a stable haemodynamic condition for at least 20 min., baseline values of diastolic blood pressure (a more accurate indicator of peripheral vascular resistance) and heart rate were determined. Subsequently, the animals (divided into sets 1 and 2; see above) underwent the following experimental protocols.

Protocol I. Electrical stimulation of the perivascular (vasodepressor) sensory out-flow. In the first set of rats (n = 93), the pithing rod was replaced by an electrode enamelled except for 1.5 cm length 9 cm from the tip, placing the uncovered segment at T_9–T_{12} of the spinal cord, and an indifferent electrode was placed dorsally [9,11–13]. Before electrical stimulation, the animals received (i.v.): (i) a bolus injection of gallamine (25 mg/kg) to avoid the electrically induced muscular twitching; (ii) 10 min. later, a continuous infusion of hexamethonium (2 mg/kg min.) to block the electrically induced vasopressor responses produced by stimulation of the pre-ganglionic sympathetic out-flow [9,11–13]; and (iii) after 10 min., a continuous infusion of methoxamine (20 μg/kg min.) to produce a sustained increase in diastolic blood pressure (at around 135 mmHg; see Results section). Then, this set of rats was divided into three groups (n = 53, 20 and 20).

The first group (n = 53) was subdivided into eleven subgroups that received an i.v. continuous infusion of (i) nothing (control; n = 5); (ii) saline (0.02 ml/min.; n = 5); (iii) 0.1, 0.3 and 1 μg/kg min. SKF-38393 (n = 4, 4 and 5, respectively); or (iv) 0.03, 0.1, 0.3, 1, 3 and 10 μg/kg min. quinpirole (n = 5 each). Twenty minutes later, the perivascular sensory out-flow was electrically stimulated during the above continuous infusions to elicit vasodepressor responses by applying 10-sec. trains of monophasic, rectangular pulses (2 ms, 50 V), at increasing frequencies of stimulation (0.56, 1, 1.8, 3.1 and 5.6 Hz). When diastolic blood pressure returned to baseline levels, the next frequency was applied. This procedure was performed until the S-R curve was completed.

The second group (n = 20) received an i.v. continuous infusion of saline (0.02 ml/min.; quinpirole vehicle) and, 10 min. later, was subdivided into four subgroups (n = 5 each) comprising i.v. bolus injections of, respectively, (i) saline (raclopride vehicle; 1 ml/kg); (ii) 5% ascorbic acid (haloperidol vehicle; 1 ml/kg); (iii) raclopride (300 μg/kg); and (iv) haloperidol (300 μg/kg). After 10 min., a S-R curve was constructed as described above to determine the effect of these antagonists *per se*.

The third group (n = 20) received an i.v. continuous infusion of quinpirole (0.1 μg/kg min.) and, after 10 min., was subdivided into four subgroups (n = 5 each) that received an i.v. bolus injection of, respectively, (i) saline (1 ml/kg); (ii) 5% ascorbic acid (1 ml/kg); (iii) raclopride (300 μg/kg); and (iv) haloperidol (300 μg/kg). After 10 min., a S-R curve was constructed as described above during the infusion of quinpirole.

It is noteworthy that only one S-R curve was carried out per animal as tachyphylaxis of the vasodepressor responses was observed when eliciting a second S-R curve, as previously reported [11–13].

Protocol II. Administration of exogenous α-CGRP. The second set of rats (n = 15) was prepared as described above, but (i) the pithing rod was left throughout the experiment (it was not replaced by the aforesaid electrode as there was no electrical stimulation) and (ii) the administration of both gallamine and hexamethonium was omitted. Then, the animals received an i.v. continuous infusion of methoxamine (20 μg/kg min.) and, 10 min. later (when diastolic blood pressure was constantly maintained at around 135 mmHg), this set was divided into three groups (n = 5 each) that received i.v. bolus injections of, respectively, (i) nothing (control group); (ii) saline (1 ml/kg); and (iii) 0.1 μg/kg min. quinpirole. Twenty minutes later, baseline values of diastolic blood pressure and heart rate were determined in the three groups as described above; subsequently, the vasodepressor responses elicited by i.v. bolus injections of exogenous α-CGRP (0.1, 0.18, 0.31, 0.56 and 1 μg/kg) were examined *during* the continuous infusion of quinpirole.

Other procedures applying to protocols I and II. Hexamethonium, methoxamine and the dopaminergic agonists were continuously infused at a rate of 0.02 ml/min. Diastolic blood pressure was increased in all cases with the methoxamine infusion. The interval between the different frequencies of stimulation and doses of α-CGRP depended on the duration of the resulting vasodepressor responses (5–10 min.), as we waited until diastolic blood pressure had returned to baseline values.

Compounds. Apart from the anaesthetic (diethyl ether), the compounds used in this study (obtained from the sources indicated) were as follows: gallamine triethiodide, hexamethonium chloride, rat α-CGRP, methoxamine hydrochloride, quinpirole dihydrochloride, (±)-1-phenyl-2,3,4,5-tetrahydro-(1*H*)-3-benzazepine-7,8-diol hydrochlo ride (SKF-38393), raclopride tartrate (Sigma Chemical Co., St. Louis, MO, USA), and haloperidol (gift: Janssen-Cilag, Mexico City, Mexico). All compounds were dissolved in saline, except haloperidol, which was dissolved in 5% (w/v) ascorbic acid in saline; these vehicles had no effect on the baseline values of diastolic blood pressure and heart rate (not shown).

The doses mentioned in the text refer to the salts of substances, except in the case of SKF-38393, quinpirole and haloperidol, where they refer to the free base.

Data presentation and statistical evaluation. All data in the text and figures are presented as means ± S.E.M. The changes in diastolic blood pressure produced by either electrical stimulation or exogenous α-CGRP in saline- and agonist-infused animals were determined and expressed as per cent change from baseline at the steady-state effect of methoxamine. The difference between the changes in diastolic blood pressure induced by electrical stimulation and exogenous α-CGRP within the different subgroups of animals was compared with a two-way analysis of variance. Such analysis was followed, if applicable, by the Student–Newman–Keuls *post hoc* test. Statistical significance was accepted at $p < 0.05$.

Results

Systemic haemodynamic variables. The baseline values of diastolic blood pressure and heart rate in the 108 rats were 68 ± 2 mmHg and 286 ± 5 beats/min., respectively. These values remained unchanged after the administration of gallamine and hexamethonium (not shown) in the animals receiving thoracic (T_9–T_{12}) electrical stimulation (n = 93, see below). As expected, 20 min. after starting the infusion of methoxamine, diastolic blood pressure was significantly increased (139 ± 3 mmHg; $p < 0.05$), with heart rate remaining practically unchanged. This sustained vasopressor response remained unaltered during the continuous infusions of saline (0.02 ml/min.), SKF-38393 (0.1–1 μg/kg min.) or quinpirole (0.03–10 μg/kg min.), as well as after i.v. administration of raclopride.

In contrast, the i.v. administration of haloperidol induced a significant, but transient, decrease in diastolic blood pressure and heart rate; both variables returned to baseline values after 10 min. (data not shown).

Vasodepressor responses elicited by electrical stimulation or exogenous α-CGRP. The stimulation of the perivascular sensory vasodepressor out-flow (0.56–5.6 Hz) and i.v. bolus injections of exogenous α-CGRP (0.1–1 μg/kg) resulted in, respectively, frequency-dependent and dose-dependent vasodepressor responses (figs 1A and 2A), which were immediate in onset. In accordance with previous findings [9,11–13], the above vasodepressor responses (i) appeared about 10 sec. after starting each electrical or pharmacological stimulus; (ii) reached a maximum 1 min. after the stimulus had ended; and (iii) returned to baseline values after 5–10 min. These responses were due to selective vasodepressor stimulation, as only negligible and inconsistent effects in heart rate were observed (not shown), as previously reported [9,11–13]. It is noteworthy that (i) this response profile was also produced during the infusion of SKF-38393 and quinpirole (not shown) and (ii) Villalón *et al.* [11] have previously shown that during the infusion of a greater dose of methoxamine (i.e. 30 μg/kg min., which produced a higher and sustained increase in diastolic blood pressure), electrical stimulation produced similar ($p > 0.05$) frequency-dependent vasodepressor responses.

Vasodepressor responses elicited by electrical stimulation or exogenous α-CGRP during the infusions of saline, SKF-38393 or quinpirole. The electrically induced vasodepressor responses were (i) unaltered during the infusions of saline (0.02 ml/min., fig. 1A), SKF-38393 (0.1–1 μg/kg min.; fig. 1B) or 0.03 μg/kg min. quinpirole (fig. 1C) and (ii) significantly inhibited (particularly at 1.8, 3.1 and 5.6 Hz) during the infusions of 0.1–10 μg/kg min. quinpirole (fig. 1C, D). As 0.1 μg/kg min. quinpirole produced a marked inhibition that did not significantly differ ($p > 0.05$) from that induced by 0.3, 1, 3 and 10 μg/kg min. quinpirole (fig. 1C, D), 0.1 μg/kg min. quinpirole was chosen for further pharmacological analysis.

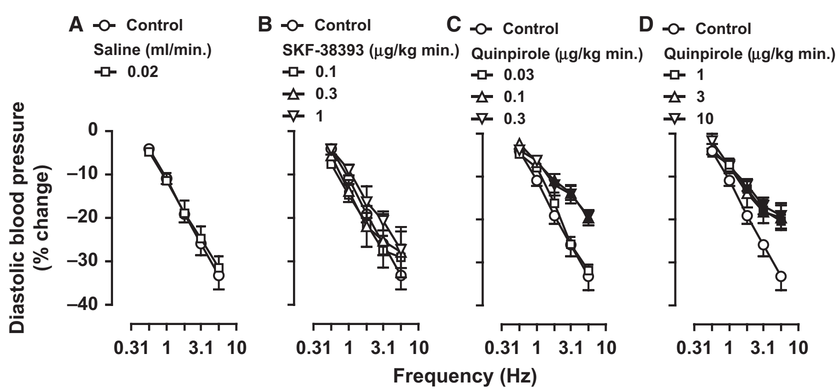

Fig. 1. Vasodepressor responses induced by electrical stimulation elicited during i.v. continuous infusions of saline (A: 0.02 ml/min.), SKF-38393 (B: 0.1, 0.3 and 1 μg/kg min.) or quinpirole (C: 0.03, 0.1 and 0.3 μg/kg min.; and D: 1, 3 and 10 μg/kg min.). Empty symbols depict either control responses (○) or non-significant responses (□△▽) *versus* control. Solid symbols (■▲▼) represent significantly different responses ($p < 0.05$) *versus* control.

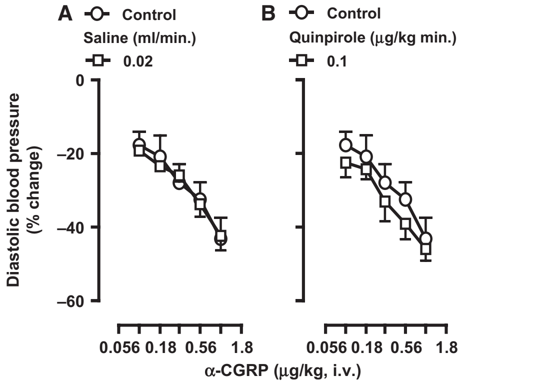

Fig. 2. Vasodepressor responses to i.v. bolus injections of α-CGRP elicited during i.v. continuous infusions of vehicle (A: saline; 0.02 ml/min.) or quinpirole (B: 0.1 μg/kg min.). Empty symbols depict either control responses (○) or non-significant responses (□) *versus* control.

Indeed, the vasodepressor responses to α-CGRP, which remained unaltered during the infusion of saline (fig. 2A), were not affected by 0.1 μg/kg min. quinpirole (fig. 2B). Moreover, as the vasodepressor responses to electrical stimulation remained unaltered during the infusions of the D_1-like receptor agonist SKF-38393 (fig. 1B), the vasodepressor responses to α-CGRP were not analysed during SKF-38393 infusions.

Effect of vehicles, raclopride or haloperidol per se on the electrically induced vasodepressor responses.. Figure 3 illustrates that the vasodepressor responses induced by electrical stimulation in control animals were not significantly different ($p > 0.05$) from those elicited in animals pre-treated with an i.v. bolus injection of saline (1 ml/kg; fig. 3A), 5% ascorbic acid (1 ml/kg; fig. 3B) or raclopride (300 μg/kg; fig. 3C), during an i.v. continuous infusion of saline (0.02 ml/min.). Similar results were obtained in the animals pre-treated with haloperidol (300 μg/kg), except that the vasodepressor response induced by 5.6 Hz was significantly attenuated (fig. 3D).

Effect of vehicles, raclopride or haloperidol on quinpirole-induced inhibition of the electrically induced vasodepressor responses.. Figure 4 shows that the inhibition of the electrically induced vasodepressor responses induced by 0.1 μg/kg min. quinpirole was (i) unchanged in animals pre-treated with saline (1 ml/kg; fig. 4A) or 5% ascorbic acid (1 ml/kg; fig. 4B) and (ii) abolished in animals pre-treated

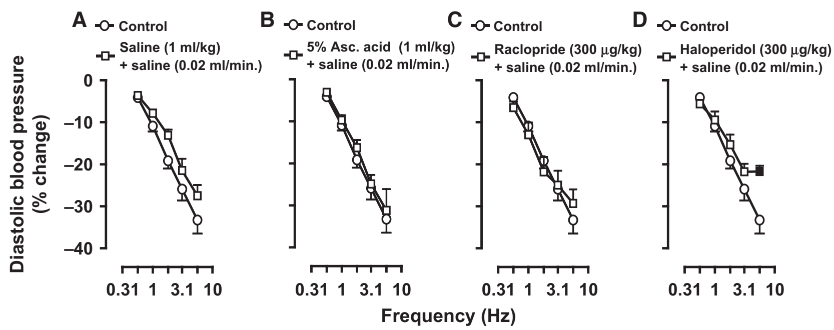

Fig. 3. Effect of i.v. bolus injections of saline (A: 1 ml/kg), 5% ascorbic acid (5% Asc. acid; B: 1 ml/kg), raclopride (C: 300 μg/kg) or haloperidol (D: 300 μg/kg) *per* se on the vasodepressor responses induced by electrical stimulation during an i.v. continuous infusion of saline (0.02 ml/min.). Empty symbols depict either control responses (○) or non-significant responses (□) *versus* control. Solid symbols (■) represent significantly different responses ($p < 0.05$) *versus* control. The control group is the same as that depicted in fig. 1A, but it is shown for the sake of clarity when making comparisons.

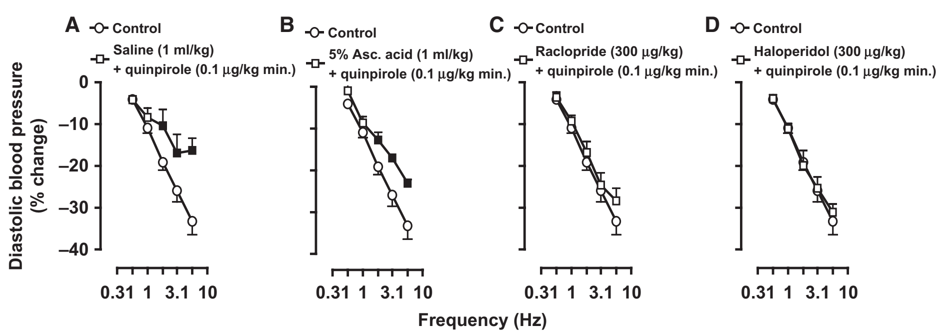

Fig. 4. Effect of i.v. bolus injections of saline (A: 1 ml/kg), 5% ascorbic acid (5% Asc. acid; B: 1 ml/kg), raclopride (C: 300 μg/kg) or haloperidol (D: 300 μg/kg) on the inhibition by quinpirole (0.1 μg/kg min.) of the electrically induced vasodepressor responses. For the sake of clarity, the control group is shown for comparison. Empty symbols depict either control responses (○) or non-significant responses (□) *versus* control. Solid symbols (■) represent significantly different responses ($p < 0.05$) *versus* control.

with 300 μg/kg of either raclopride (fig. 4C) or haloperidol (fig. 4D).

Discussion

General. Through the years, numerous groups have reported that activation of pre-junctional D_2-like receptors produces sympatho-inhibition in several experimental models [1–4], including the systemic vasculature of pithed rats [5]. Our study now shows, for the first time, the role of D_2-like receptors inhibiting the vasodepressor sensory CGRPergic out-flow in pithed rats. In this respect, perivascular sensory nerve activity (i.e. electrically induced CGRP release) was estimated indirectly by measurement of the evoked vasodepressor response, as demonstrated earlier [9,11–13]. So, the responses to quinpirole were considered sensory inhibitory (via D_2-like receptors; see below) as this agonist (at 0.1 μg/kg min.) inhibited the vasodepressor responses to stimulation of the sensory CGRPergic out-flow (fig. 1C), but not those to exogenous α-CGRP (fig. 2B).

Moreover, it is noteworthy that only one S-R curve was carried out per animal under our experimental conditions, as tachyphylaxis of the vasodepressor responses was observed when eliciting a second S-R curve, as previously reported by our group [11–13]. The mechanisms involved in this tachyphylactic phenomenon may include (but are not limited to) depletion of neuronal CGRP, uncoupling from the G protein and/or receptor and down-regulation of CGRP receptors; notwithstanding, the extent to which each of these mechanisms is involved was not assessed in our study.

Interestingly, 0.1 and 0.3 μg/kg min. quinpirole produced a similar inhibition of the vasodepressor sensory CGRPergic out-flow (fig. 1C), whereas 1, 3 and 10 μg/kg min. quinpirole produced no greater inhibition (fig. 1D). While we have no explanation for this finding, the fact that the inhibition by quinpirole was abolished by 300 μg/kg of the D_2-like receptor antagonists raclopride (fig. 4C) or haloperidol (fig. 4D) suggests the role of D_2-like receptors which, upon activation, inhibit the vasodepressor sensory out-flow.

Systemic haemodynamic effects produced by the different treatments. As previously established [9,11–13], stimulation of the perivascular sensory CGRPergic out-flow produces vasodepressor responses in pithed rats only when diastolic blood pressure was maintained at around 135 mmHg by a methoxamine infusion. This sustained vasopressor response by methoxamine results from an increase in peripheral vascular resistance due to stimulation of vascular α_1-adrenoceptors [19].

It must be highlighted that the selective agonists SKF-38393 (D_1-like) or quinpirole (D_2-like) did not affect the increase in diastolic blood pressure during the infusion of methoxamine in our experiments (not shown). In contrast, other studies have shown that activation of these receptors results in a dose-dependent decrease in diastolic blood pressure [20,21]. This apparent discrepancy could be due to the fact that i.v. continuous infusions (in low doses) of these agonists were used in our study (fig. 1). Indeed, even higher doses of these compounds (as i.v. continuous infusions) in the pithed rat model failed to modify the baseline values of diastolic blood pressure or heart rate [5].

Furthermore, the fact that haloperidol induced a transient hypotension and that it attenuated the vasodepressor response induced by 5.6 Hz (fig. 3D) most likely reflects a non-specific effect as it also displays affinity for other receptors, including α_1-adrenoceptors and 5-HT_2 receptors [22,23]. Clearly, haloperidol is less selective than raclopride as a D_2-like receptor antagonist (table 1); however, we still decided to use haloperidol here since 20 years ago one study reported that it blocked dopamine-induced inhibition of the NANC neurotransmission in the isolated rat bladder [14].

Evidence supporting the role of D_2-like, rather than D_1-like, receptors. The dopamine receptors that inhibit the vasodepressor sensory CGRPergic out-flow in our study closely resemble the pharmacological profile of the D_2-like (rather than the D_1-like) type for several reasons. In the first instance, the inhibition induced by quinpirole (a D_2-like receptor agonist; table 1) on the electrically induced vasodepressor responses, in doses as low as 0.1 μg/kg min. (fig. 1C), suggests the involvement of D_2-like receptors. This suggestion is clearly reinforced by the capability of 300 μg/kg of either raclopride or haloperidol (D_2-like receptor antagonists; table 1) to abolish the inhibition induced by quinpirole (fig. 4C,D); this dose of raclopride or haloperidol is high enough to block pre-junctional D_2-like receptors in pithed rats [5]. Consistent with the above findings, SKF-38393 (a potent and selective D_1-like receptor agonist; table 1) failed to inhibit the electrically induced vasodepressor responses in doses up to 1 μg/kg min. (fig. 1B). Regarding the specific D_2-like receptor subtypes (i.e. D_2, D_3 and D_4) involved in the above inhibition by quinpirole, it is noteworthy that (i) haloperidol cannot distinguish among these subtypes (table 1) and (ii) the highest affinities of raclopride for the D_2 and D_3 subtypes (table 1) might suggest that either (or both) subtypes are involved. Clearly, further experiments will be required to ascertain the specific D_2-like receptor subtypes involved in the above inhibition by quinpirole.

Table 1.
Binding affinity constants (pK_i) of several agonists and antagonists for dopamine receptors.

	D_1-like		D_2-like		
	D_1	D_5	D_2	D_3	D_4
SKF-38393[1]	9.0	~9.3	~6.8	~5.3	~6.0
Quinpirole[1]	5.7	–	8.3	~7.6	~7.5
Raclopride	4.7	–	8.7	8.4	5.6
Haloperidol	~7.1	~7.0	8.9	~8.2	8.6

Data taken (as K_i) from Seeman and Van Tol [6].
~, stands for 'approximately'.
[1]The corresponding K_i values for SKF-38393 and quinpirole were determined at the high-affinity state for all receptors, except for the D_3 receptor.

Admittedly, our study provides no direct transductional evidence that the inhibition produced by quinpirole involves activation of a receptor-coupled $G_{i/o}$ proteins in sensory neurons. Nevertheless, D_2-like receptors are, by definition, coupled to $G_{i/o}$ proteins that inhibit adenylyl cyclase activity, inactivate Ca^{2+} channels and/or activate inwardly rectifying K^+ channels [1,2,6]. These are signal transduction systems usually associated with inhibition of neurotransmitters release, while the opposite (i.e. facilitation of neurotransmitters release) applies for D_1-like receptors [24]. Interestingly, the fact that both raclopride and haloperidol failed to potentiate the electrically induced vasodepressor responses (fig. 3C,D) suggests that D_2-like receptors do not play a predominant physiological role in the modulation of perivascular sensory CGRP release.

Possible locus of the sensory inhibitory D_2-like receptors. Lastly, one might speculate upon the locus of the D_2-like receptors inhibiting the vasodepressor sensory CGRPergic out-flow. In the first instance, Taguchi *et al.* [9] demonstrated that the electrically induced vasodepressor responses are mainly mediated by CGRP release. Consequently, we assume that quinpirole-induced inhibition of the vasodepressor sensory CGRPergic out-flow (via D_2-like receptors) may actually involve inhibition of CGRP release from sensory nerves. However, our study provides no direct evidence to support this view (e.g. measurement of plasma CGRP levels during quinpirole infusions). Moreover, central or spinal mechanisms can be excluded as pithed rats were used. Hence, the sensory inhibitory D_2-like receptors could be located on sensory perivascular nerves and/or dorsal root ganglia as quinpirole failed to inhibit the vasodepressor responses induced by exogenous α-CGRP (fig. 2B). Indeed, other findings obtained by molecular biology techniques have shown the presence of mRNAs for all dopamine receptor subtypes in rat dorsal root ganglia neurons [25]. However, the specific cell population expressing these subtypes remains elusive.

In conclusion, the above results, taken together, suggest that the inhibition produced by 0.1 μg/kg min. quinpirole on the vasodepressor sensory CGRPergic out-flow is mainly mediated by pre-junctional D_2-like receptors.

Acknowledgements

The authors would like to thank Mr. Mauricio Villasana and Q.F.B. Belinda Villanueva for their skilful technical assistance. This study was financially supported by Consejo Nacional de Ciencia y Tecnología (CONACyT, Project No. 60789; México D.F.).

Conflict of Interest

The authors declare that they have no conflict of interest.

References

1 Missale C, Nash S, Robinson S, Jaber M, Caron M. Dopamine receptors: from structure to function. Physiol Rev 1998;**78**:189–225.

2 Beaulieu JM, Gainetdinov RR. The physiology, signaling, and pharmacology of dopamine receptors. Pharmacol Rev 2011;**63**:182–217.

3 Goldberg LI. Cardiovascular and renal actions of dopamine: Potential clinical applications. Pharmacol Rev 1972;**24**:1–24.

4 Goldberg LI, Volkman PH, Kohli JD. A comparison of the vascular dopamine receptors with other dopamine receptors. Annu Rev Pharmacol Toxicol 1978;**18**:57–79.

5 Manrique-Maldonado G, González-Hernández A, Marichal-Cancino BA, Villamil-Hernández MT, del Mercado OA, Centurión D *et al.* The dopamine receptors mediating inhibition of the sympathetic vasopressor outflow in pithed rats: pharmacological correlation with the D(2)-like type. Basic Clin Pharmacol Toxicol 2011;**109**:506–12.

6 Seeman P, Van Tol MH. Dopamine receptor pharmacology. Trends Pharmacol Sci 1994;**15**:264–70.

7 Jose PA, Eisner GM, Felder RA. Regulation of blood pressure by dopamine receptors. Nephron Physiol 2003;**95**:19–27.

8 Hoffman BB. Catecholamines, sympathomimetic drugs, and adrenergic receptor antagonists. In: Hardman JG, Limbird LE, Goodman Gilman A (eds). Goodman & Gilman's The Pharmacological Basis of Therapeutics, 10th edn. McGraw-Hill, New York, 2001;215–68.

9 Taguchi T, Kawasaki H, Imamura T, Takasaki K. Endogenous calcitonin gene–related peptide mediates nonadrenergic noncholinergic depressor response to spinal cord stimulation in the pithed rat. Circ Res 1992;**71**:357–64.

10 Van Rossum D, Hanisch UK, Quirion R. Neuroanatomical localization, pharmacological characterization and functions of CGRP, related peptides and their receptors. Neurosci Biobehav Rev 1997;**21**:649–78.

11 Villalón CM, Albarrán-Juárez JA, Lozano-Cuenca J, Pertz HH, Görnemann T, Centurión D. Pharmacological profile of the clonidine-induced inhibition of vasodepressor sensory outflow in pithed rats: correlation with alpha(2A/2C)–adrenoceptors. Br J Pharmacol 2008;**154**:51–9.

12 González-Hernández A, Muñoz-Islas E, Lozano-Cuenca J, Ramőrez-Rosas MB, Sánchez-López A, Centurión D *et al.* Activation of 5-HT1B receptors inhibits the vasodepressor sensory CGRPergic outflow in pithed rats. Eur J Pharmacol 2010;**637**:131–7.

13 González-Hernández A, Manrique-Maldonado G, Lozano-Cuenca J, Muñoz-Islas E, Centurión D, Maassen VanDenBrink A *et al.* The 5-HT(1) receptors inhibiting the rat vasodepressor sensory CGRPergic outflow: further involvement of 5-HT(1F), but not 5-HT(1A) or 5-HT(1D), subtypes. Eur J Pharmacol 2011;**659**:233–43.

14 Lot TY. Dopamine depresses non-adrenergic, non-cholinergic neurotransmission in the rat bladder. J Pharm Pharmacol 1993;**45**:570–2.

15 Bayne K. Revised Guide for the Care and Use of Laboratory Animals available. American Physiological Society. Physiologist 1996;**39**:199.

16 McGrath JC, Drummond GB, McLachlan EM, Kilkenny C, Wainwright CL. Guidelines for reporting experiments involving animals: the ARRIVE guidelines. Br J Pharmacol 2010;**160**:1573–6.

17 Gillespie JS, MacLaren A, Pollock D. A method of stimulating different segments of the autonomic outflow from the spinal column to various organs in the pithed cat and rat. Br J Pharmacol 1970;**40**:257–67.

18 Kleiman LI, Radford EP. Ventilation standards for small mammals. J Appl Physiol 1964;**19**:360–2.

19 Decker N, Ehrhardt JD, Leclerc G, Schwartz J. Postjunctional alpha-adrenoceptors. Alpha 1 and alpha 2 subtypes in rat vasculature in vitro and in vivo. Naunyn- Schmiedeberg's Arch Pharmacol 1984;**326**:1–6.

20 Luippold G, Max A, Albinus M, Osswald H, Mühlbauer B. Role of the renin-angiotensin system in the compensation of

quinpirole-induced blood pressure decrease. Naunyn-Schmiedeberg's Arch Pharmacol 2003;**367**:427–33.

21 Ventura HO, Messerli FH, Frohlich ED, Kobrin I, Oigman W, Dunn FG *et al*. Immediate hemodynamic effects of a dopamine-receptor agonist (fenoldopam) in patients with essential hypertension. Circulation 1984;**69**:1142–5.

22 Leysen JE. Serotonergic binding sites. In: Vanhoutte PM (ed.). Serotonin and the Cardiovascular System. Raven Press, New York, 1985;43–62.

23 D'Souza DC, Braley G, Blaise R, Vendetti M, Oliver S, Pittman B *et al*. Effects of haloperidol on the behavioral, subjective, cognitive, motor, and neuroendocrine effects of Delta-9-tetrahydrocannabinol in humans. Psychopharmacology 2008;**198**:587–603.

24 De Jong AP, Verhage M. Presynaptic signal transduction pathways that modulate synaptic transmission. Curr Opin Neurobiol 2009;**19**:245–54.

25 Xie GX, Jones K, Peroutka SJ, Palmer PP. Detection of mRNAs and alternatively spliced transcripts of dopamine receptors in rat peripheral sensory and sympathetic ganglia. Brain Res 1998;**785**:129–35.